Chapter-8

CIRCULAR MOTION

- If a force is always $\perp^r$ to the direction of motion (i.e. velocity) then path will be **circular.**
- In circular motion vector quantities like velocity, linear momentum, acceleration, force changes.

Uniform circular motion

(i) Speed (v) is constant

(ii) 'ω' (angular velocity) is constant

(iii) KE is constant,

(iv) $\Delta\omega = 0, \Delta k = 0$

(v) Work done and power is zero

(vi) Tangential acceleration (Rate of change of speed)

$$a_t = \frac{d|\vec{v}|}{dt} = 0$$

(vii) $a_t = r\,\alpha = 0 \Rightarrow \alpha = 0$

(viii) Acceleration is directed along radius toward centre known as centripetal acceleration or radial accl^n or Normal accl^n.

$$a_c = \frac{v^2}{r} = \omega^2 r = 4\pi^2 f^2 r = \frac{4\pi^2}{T^2} r$$

(ix) Total accl^n is only due to centripetal accl^n

(x) Angular momentum (L) is constant

(xi) $\tau = \frac{dL}{dt} = 0$

(xii) KE, $k = \frac{1}{2} mv^2 = \frac{1}{2} F_C \times r$

(xiii) Total energy, $TE = \frac{1}{2} F_C \times r + \int -F_C \, .dr$

Non-uniform circular motion:

(i) Speed changes (ii) ω changes (iii) KE changes

(iv) $a_t \neq 0, \Delta\omega \neq 0, \Delta k \neq 0, \alpha \neq 0$

(v) If speed increases then tangential acceleration (a_t) is along the direction of velocity.

i.e. Angle between $\vec{a_t}$ and $\vec{v}$ is 0°

$$\vec{a_t} \cdot \vec{v} = +ve$$

(vi) If speed decreases then tangential acceleration is opposite in the direction of velocity.

$$\vec{a_t} \cdot \vec{v} = -ve$$

(vii) Total acceleration is due to both centripetal and tangential

$$a_{Total} \text{ or } a = \sqrt{a_c^2 + a_t^2} = \sqrt{\left(\frac{v^2}{r}\right)^2 + \left(\frac{dv}{dt}\right)^2}$$

- Direction of total accl^n from a_c, $\tan\beta = \frac{a_t}{a_c}$

(viii) The work done power both are due to tangential force.

(ix) Angular momentum changes, $L = m\,v\,r$

(x) Torque $\neq 0$, $\tau = m\,a_t\,r$, Torque is due to tangential force.

- Work done by centripetal force is always zero in circular motion

For motion of cyclist, Angle made by cyclist with vertical

$$\tan\theta = \frac{v^2}{rg}$$

$\Rightarrow$ If the length of circumference of circular path is given then

$$\tan\theta = \frac{2\pi c}{t^2 g}$$

where,

c = length of circumference

t = time taken to complete circular path

For banking of roads

$$\tan\theta = \frac{v^2}{rg}$$

Height raised by outer edge w.r.to inner edge (super elevation)

$$h = \frac{v^2}{rg} \times L \text{ where, } L = \text{width of road,}$$

- Max^m speed of which not for skidding on level Curved road (i.e. curved road of without banking)

$$v_{max} = \sqrt{\mu rg}$$

where, μ = coefficient of friction between tyre and road,

- Max^m speed of vehicle on such road not for overturning

$$v_{max} = \sqrt{\frac{rdg}{2h}}$$

where,

d = distance between wheels

h = height of cg of vehicle from ground

- If banking and roughness both are provided then min^m and max^m speed of vehicle not for skidding down and up.

  $v_{min} = \sqrt{rg\tan(\theta - \alpha)}$,

  where θ = banking angle

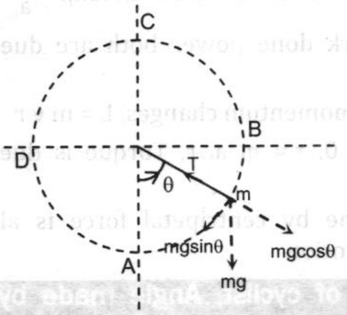
  $v_{max} = \sqrt{rg\tan(\theta + \alpha)}$

  α = angle of friction[$\tan\alpha = \mu$]

Motion in vertical circle:

ℓ = length of string

or Radius of circle (r)



- If body of mass 'm' is tied to string of length 'ℓ' and is rotated in vertical circle then from lowest point to highest point **both tension and velocity** decreases.
- Tension in the string at any angle 'θ' with vertical from lowest position.

  $T = mg\cos\theta + \frac{m v^2}{r}$

  For lowest point, θ = 0°, $T_A = mg + \frac{mv_4^2}{r} = T_{max}$

  For highest point, θ = 180°, $T_C = \frac{mr_c^2}{r} - mg = T_{min}$
- When string become horizontal, θ = 90° or 270°

  $T = \frac{mv^2}{r}$
- If velocity of body at lowest point is 'v' then velocity at any height,

  $v_h = \sqrt{v^2 - 2gh}$

  At any angle 'θ' height is, $h = \ell(1 - \cos\theta)$
- velocity when string becomes horizontal, θ = 90° or, h = ℓ

  $v_h = \sqrt{v^2 - 2g\ell}$
- Change in velocity when string becomes horizontal

  $|\overrightarrow{\Delta v}| = \sqrt{v_h^2 + v_A^2}$

  $|\overrightarrow{\Delta v}| = \sqrt{2(v^2 - g\ell)}$
- Difference between max^m tension and min^m tension.

  $T_{max} - T_{min} = 6mg$
- Ratio of max^m tension to min^m tension

  $\frac{T_{max}}{T_{min}} = \frac{v_A^2 + rg}{v_A^2 - 5rg} = \frac{v_C^2 + 5rg}{v_C^2 - rg}$

Critical velocity at different points

- To complete vertical circle tension in string at highest point $T_c \geq 0$.

  $(V_C)_{min} = \sqrt{rg}$; $T_c = 0$

  $V_B = V_D = v_{max} = \sqrt{3rg}$; $T = 3mg$

  $(V_A)_{max} = \sqrt{5rg}$, $T = 6mg$

- Max^m speed of vehicle moving on **Hump bridge** (Convex bridge) of radius 'r' not for losing contact.

  $v_{max} = \sqrt{rg}$
- Time period rotation to complete vertical circle of radius 'r'

  $T = 2\pi\sqrt{\frac{r}{g}} = 2\sqrt{r}$ sec
- Min^m height from which a body slides down on frictionless track to complete a circular loop of radius 'r' at bottom (or end).

  $h_{min} = \frac{5r}{2} = \frac{5D}{4}$ where D = diameter
- If a body is tied to string of length 'ℓ' is given horizontal velocity v_A at lowest point such that

(i) If $V_A < \sqrt{2g\ell}$

In this case velocity becomes zero earlier than tension and body oscillates.

- Max^m height attained by body; $h = \frac{v_A^2}{2g} < \ell$
- Angle made by string with vertical

  $\cos\theta = 1 - \frac{v_A^2}{2g\ell}$

(ii) If $\sqrt{2g\ell} < v_A < \sqrt{5g\ell}$,

In this case tension becomes zero earlier than velocity. In this case, body leaves the circular path and moves under gravity in parabolic path.

- The height from lowest point at which body leaves the path, $h = \frac{v_A^2 + g\ell}{3g}$
- Angle with vertical (from high)

  $\cos\theta = \frac{v_A^2 - 2g\ell}{3g\ell}$
- Velocity at that point $v = \sqrt{\frac{v_A^2 - 2g\ell}{3}}$
- Max^m height attained body from lowest point

  $H = \frac{v_A^2 + g\ell}{3g} + \frac{v^2\sin^2\theta}{2g} < 2\ell$
- If a body is released from top of hemispherical bowl o radius 'R" then body leaves the bowl

  (i) at height $h = \frac{2R}{3}$ from centre

  (ii) at $h = \frac{R}{3}$ from top

  (iii) at $\cos\theta = \frac{2}{3}$; (θ = 48°) with vertical

- In case of spherical bowl height from bottom = $\frac{5R}{3}$
- If a vehicle is moving on concave track and convex track then normal force exerted by concave track is always greater than convex track.
- If a vehicle is moving on convex track of different radii then that curvature exert more force for which radius is larger.
- If case of concave track smaller curvature exert more force.
- If a uniform rod of mass m_1 having length 'L' is rotated about one end with angular velocity 'ω' then force exerted on the another end.

$$F = \frac{1}{2} m\, \omega^2 L$$

- If a body of mass 'm' is tied to spring of natural length (or relaxed length) 'ℓ' of force constant 'k' is rotated with angular velocity 'ω' then elongation in the spring

$$x = \frac{m\, \omega^2 \ell}{k - m\omega^2}$$

The max^m distance from centre at which a coin may be placed on a circular table rotating with angular velocity 'ω' about an axis through cg and $\perp$r to plane.

$$x_{max} = \frac{\mu g}{\omega^2} \quad \text{where } \mu = \text{coefficient friction}$$

- If a circular table is rotating from rest with constant angular acceleration 'α' and a coin is placed at a distance x from centre then it starts slipping after

(i) The min^m - time, $t = \sqrt{\frac{\mu g}{\alpha^2 x}}$

(ii) Min^m No. of revolution (n) $= \frac{\theta}{2\pi} = \frac{\mu g}{4\pi\, \alpha\, x}$

Conical pendulum:

(i) Tension in the string: $T = \frac{mg}{\cos\theta}$ [where, $\sin\theta = \frac{r}{l}$]

(ii) For linear speed of bob, $\tan\theta = \frac{v^2}{rg}$

(iii) Time period, $T_c = 2\pi\sqrt{\frac{\ell \cos\theta}{g}} = T_s \times \sqrt{\cos\theta}$

$$\frac{T_c}{T_s} < 1 : \frac{T_c}{T_s} = \left(1 - \frac{r^2}{\ell^2}\right)^{\frac{1}{4}}$$

where,

r = radius of horizontal circle

ℓ = length of string

T_s = Time period of simple pendulum

Multiple Choice Questions

1. A wheel is at rest. Its angular velocity increases uniformly and becomes 80 radian per second after 5 second. The total angular displacement is:

a. 800 rad (b) 400 rad
c. 200 rad d. 100 rad

2. A particle of mass m begins to slide down a fixed smooth sphere from the top. what is its tangential acceleration when it breaks off the sphere?

a. $\frac{2g}{3}$ b. $\frac{\sqrt{5}g}{3}$
c. g d. $\frac{g}{3}$

3. Let a_r and a_t represent radial and tangential acceleration. The motion of a particle may be circular if:

a. $a_r = 0, a_t = 0$ b. $a_r = 0, a_t \neq 0$
c. $a_r \neq 0, a_t = 0$ d. none of these

4. A particle is moving along a circular path of radius 5m with a uniform speed 5 ms^{-1}. What will be the average acceleration when the particle completes half revolution?

a. zero (b) 10 ms^{-2}
c. 10π ms^{-2} d. $\frac{10}{\pi}$ ms^{-2}

5. A 1 kg stone at the end of 1 m long string is whirled in a vertical circle at constant speed of 4m/sec. The tension in the string is 6 N when the stone is at (g = 10 m/sec^{-2}):

a. top of the circle b. bottom of the circle
c. halfway down d. none of these

6. The maximum tension that an inextensible ring of mass density 0.1 kg m^{-1} can bear is 40 N. The maximum velocity with which it can be rotated in a circular path is:

a. 20 ms^{-1} b. 18 ms^{-1}
c. 16 ms^{-1} d. 15 ms^{-1}

7. If the banking angle of a curved road is given by $\tan^{-1}\left(\frac{3}{5}\right)$ and the radius of curvature of the road is 6 m then the safe driving speed should not exceed: (g = 10 m/s^2)

a. 86.4 km/h b. 43.2 km/h
c. 21.6 km/h d. 30.4 km/h

8. A particle moving along a circular path due to a centripetal force having constant magnitude is an example of motion with:

a. constant speed and velocity
b. variable speed and velocity
c. variable speed and constant velocity
d. constant speed and variable velocity

9. A particle of mass 2 kg is moving along a circular path of radius 1m. If its angular speed is 2π rad s^{-1}, the centripetal force on it is:

a. 4π N b. 8π N
c. $4\pi^4$ N d. $8\pi^2$ N

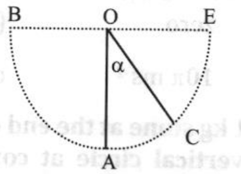
10. The speed of a particle moving in a circle is increasing. The dot product of its acceleration and velocity is:

a. negative
b. zero
c. positive
d. may be + ve or –ve

11. A particle P is moving in a circle of radius a with a uniform speed u. C is the centre of the circle and AB is a diameter. The angular velocities of P about A and C are in the ratio:

a. 1 : 1 b. 1 : 2
c. 2: 1 d. 4 : 2

12. A simple pendulum is vibrating with an angular amplitude of 90° as show in the given figure. For what value of α, is the acceleration directed?



(i) vertically upwards
(ii) horizontally
(iii) vertically downwards

a. $0°, \cos^{-1}\left(\frac{1}{\sqrt{3}}\right), 90°$

b. $90°, \cos^{-1}\left(\frac{1}{\sqrt{3}}\right), 0°$

c. $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$, 0°m 90°

d. $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right), 90°, 0°$

13. When a body moves with a constant speed along a circle:

a. its velocity remains constant
b. no force acts on it
c. no work is done on it
d. no acceleration is produced in it

14. Two racing cars of masses m_1 and m_2 are moving in circles of radii r_1 and r_2 respectively. Their speeds are such that each makes a complete circle in the same time t. The ratio of the angular speeds of the first to the second car is:

a. 1: 1 b. $m_1 : m_2$
c. $r_1 : r_2$ d. $m_1m_2 : r_1r_2$

15. A particle moves along a circle of radius $\left(\frac{20}{\pi}\right)$ m with constant tangential acceleration. If the velocity of the particle is 80 m/s at the end of the second revolution after motion has begun, the tangential acceleration is:

a. 160π m/s² b. 40π m/s²
c. 40 m/s² d. $640\,\pi$ m/s²

16. Two particles of equal mass revolving in circular paths of radii r_1 and r_2 respectively with the same angular velocity. The ratio of their centripetal force will be:

a. $\frac{r_1}{r_2}$ b. $\frac{r_2}{r_1}$

c. $\sqrt{\frac{r_2}{r_1}}$ d. $\left(\frac{r_2}{r_1}\right)^2$

17. A cyclist moving with a speed of 4.9 m/s on a level road can take a sharp circular turn of radius 4 m, then coefficient of friction between the cycle tyres and road is:

a. 0.71 b. 0.61
c. 0.31 d. 0.81

18. A bucket tied at the end of a 1.6 m long string, is whirled in a vertical circle. What should be the minimum speed so that the water from the bucket does not split when the bucket is at highest position?

a. 16 m/s b. 4 m/s
c. 6.25 m/s d. none of these

19. A body is allowed to slide down a frictionless track freely under gravity. The track end in a semicircular shaped part of a diameter D. The height (minimum) from which the body must fall so that it completes the circle, is:

a. $\frac{4}{5}$D b. $\frac{D}{2}$

c. $\frac{3D}{4}$ d. $\frac{5}{4}$D

20. A stone of mass 1 kg tied to a light inextensible string of length $L = \frac{10}{3}$ m, whirling in a circular path in a vertical plne. The ratio of maximum tension in the string to the minimum tension in the string is 4, If g is taken to be 10 m/s², the speed of the stone at the highest point of the circle is:

a. 10 m/s b. $5\sqrt{2}$ m/s
c. $10\sqrt{3}$ m/s d. 20 m/s

21. Toy cart tied to the end of an unstretched string of length a, when revolved moves in a horizonta circle of radius 2a with a time period T. Now the toy cart is speeded up until it moves in a horizontal circe of radius 3a with a period T′. If Hook's law hold then:

a. $T' = \sqrt{\frac{3}{2}}T$ b. $T' = \left(\frac{\sqrt{3}}{2}\right)T$

c. $T' = \left(\frac{3}{2}\right)T$ d. T′ = T

22.A body of mass 2 kg tied to the end of string of length 1 metre is whirled in a horizontal circle, with a uniform angular velocity of 4 rad/sec. Then, the tension of the string will be:

a. 32 N b. 16 N
c. 10 N d. 8 N

23.A motor car is travelling at 30 m/s on a circular road of radius 500 m. It is increasing its speed at the rate of 2 m/s². Then acceleration of the car will be:

a. $4\ m/s^2$ b. $3\ m/s^2$
c. $2.7\ m/s^2$ d. $8\ m/s^2$

24.A car of mass m moves in a horizontal circular path of radius metre. At an instant its speed is v m/s and is increasing at a rate a m/s², then the acceleration of the car is:

a. $a + \left(\frac{v^2}{r}\right)$ b. $\sqrt{a^2 + \left(\frac{v^2}{r}\right)^2}$
c. $\frac{v^2}{r}$ d. a

25.A small cone filled with water is revolved in a vertical circle of radius 4m and the water does not fall down. What must be the maximum period of revolution?

a. 2 sec b. 4 sec
c. 1 sec d. 6 sec

26.A cyclist goes round a circular path of circumference 34.3 m in $\sqrt{22}$ second. The angle made by him, with the vertical, will be:

a. 45° b. 90°
c. 22.5° d. 60°

27.A train has to negotiate a curve of radius 400m. By how much height should the outer rail be raised with respect to inner rail for a speed of 48 km/hr? The distance between the rails is 1m:

a. 4.5 cm b. 9 cm
c. 2.25 cm d. none of these

28.A 500 kg car takes a round turn of radius 50 m with a aped of 36 km/h. The centripetal force is:

a. 250 N b. 500 N
c. 1000 N d. 2500 N

29.The angular velocity of second hand, of a clock is:

a. $\left(\frac{\pi}{6}\right)$ rad/s b. $\left(\frac{\pi}{60}\right)$ rad/s
c. $\left(\frac{\pi}{30}\right)$ rad/s d. $\left(\frac{\pi}{15}\right)$ rad/s

30.The circular orbit of two satellites around each have radii r_1 and r_2 respectively. ($r_1 < r_2$). If angular velocities of satellite are same then their centripetal acceleration are related as:

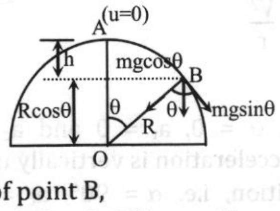
a. $a_1 > a_2$ b. $a_1 = a_2$
c. $a_1 < a_2$ d. $a_1 > a_2$

Answers:

1. c	2. b	3. c	4. d	5. a
6. a	7. c	8. d	9. d	10. c
11. b	12. a	13. c	14.a	15. c
16. a	17. b	18. b	19. d	20. a
21. b	22. a	23. c	24. b	25. b
26. a	27. a	28. c	29. c	30. c

Solution for Selected Problems

1. $w = 50$ ral/s, $\omega_0 = 0$
   $t = 5$ sec
   $\omega = \omega_0 + \alpha t \quad \Rightarrow \alpha = 16\ ral/s^2$
   $\theta = \omega_0 t + \frac{1}{2}\alpha t^2 = \frac{1}{2} \times 16 \times 5^2 = 200rad$
   Let at point 'B" particle breaks off the sphere
   $h = R - R\cos\theta = R(1 - \cos\theta)$



   Velocity of point B,
   $V_B^2 = V_A^2 + 2gh = 2gR(1 - \cos\theta)$ ....(i)
   Again, $mg\cos\theta = \frac{mV_B^2}{R} \Rightarrow V_B^2 = gR\cos\theta$ ....(ii)
   From (i) & (ii)
   $\cos\theta = \frac{2}{3}$
   Tangential force at the point = mg sinθ
   Tangent accl^n, $a_t = g\sin\theta$
   $\therefore \quad a_t = \frac{\sqrt{5}}{3} g$

5. Tension at any point making an angle 'θ' with vertical from lowest point
   $T = mg\cos\theta + \frac{mV^2}{r}$
   $6 = 1 \times 10\cos\theta + \frac{1 \times 4^2}{1}$
   $\cos\theta = -\frac{10}{10} = -1 \ \therefore \theta = 180°$
   i.e. at top of the circle

6. $Tmax = \frac{m V_{max}^2}{\ell}$
   $\Rightarrow \quad 40 = 0.1 \times V_{max}^2$
   $V_{max}^2 = 400$
   $\therefore \quad V_{max} = 20$ m/sec

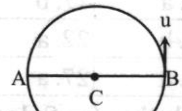
7. $\theta = \tan^{-1}\left(\frac{3}{5}\right)$
   $\therefore \quad \tan\theta = \frac{V^2}{rg}$

$V^2 = \frac{3}{5} \times 6 \times 10$

$V = 6$ m/sec

$V = \left(6 \times \frac{18}{5}\right)$ km/hr = 21.6 km/hr

11. Angular velocity, $w = \frac{V}{r}$

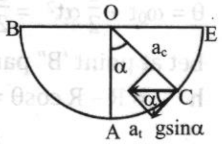


About A; $w_A = \frac{V}{2r}$

About C; $w_C = \frac{V}{r}$

$\frac{w_A}{w_C} = \frac{1}{2} = 1 : 2$

12. Tangential acceleration at any angle 'α' with vertical is given by:



$a_t = g\sin\alpha$ and centripetal acceleration, $a_c = \frac{V^2}{r}$

$\vec{a} = \vec{a_c} + \vec{a_t}$

⇒ At lowest point, $\alpha = 0$, $a_t = 0$ and a_c is vertically upwards hence acceleration is vertically upward

⇒ At extreme position, i.e. $\alpha = 90°$, $a_t = g$ vertically downward and $v = 0$, $a_c = 0$ hence total acceleration vertically downward

⇒ When total acceleration is horizontal, it makes an angle 'α' with tangential acceleration,

$a_c = \frac{v^2}{r} = \frac{2gr\cos\alpha}{r} = 2g\cos\alpha$

$\tan\alpha = \frac{a_c}{a_t} \Rightarrow \tan^2\alpha = 2$

$\therefore \tan\alpha = \sqrt{2}$

$\therefore \cos\alpha = \frac{1}{\sqrt{3}}$

$\alpha = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

14. $\omega = \frac{2\pi}{T}$

Time period for both car is same hence, $\frac{\omega_1}{\omega_2} = 1 : 1$

15. $r = \frac{20}{\pi}$, $v = 80$ m/sec

$v^2 = u^2 + 2a_t \times s = 0 + 2a_t \times s$

$\therefore a_t = \frac{80^2}{2 \times \left(2\pi \times \frac{20}{\pi}\right) \times 2} = 40\ \text{m/s}^2$

17. $v^2 = \mu rg$

$\therefore \mu = \frac{4.9^2}{4 \times 9.8} = 0.61$

18. Minimum speed at highest point,

$V = \sqrt{rg} = \sqrt{1.6 \times 10} = 4$ m/sec

19. Velocity of lowest point in terms of height

$V = \sqrt{2gh}$

To complete the vertical circle minimum velocity of lowest point

$V = \sqrt{5rg} = \sqrt{\frac{5Dg}{2}}$

$\therefore \sqrt{2gh} = \sqrt{\frac{5Dg}{2}}$

$h_{min} = \frac{5D}{4}$

20. $\frac{T_{max}}{T_{min}} = \frac{V^2 + 5rg}{V^2 - rg}$

$\Rightarrow 4 = \frac{V^2 + 5rg}{V^2 - rg}$

$\Rightarrow V^2 = 3rg = 3 \times \frac{10}{3} \times 10$

$\therefore V = 10$ m/sec

21. $k \times a = m \times \left(\frac{2\pi}{T}\right)^2 \times 2a$ ....(i)

and $k \times 2a = m \times \left(\frac{2\pi}{T'}\right)^2 \times 3a$ ....(ii)

Solving (i) and (ii) we get

$T' = \frac{\sqrt{3}}{2} T$

22. Tension in string = $mw^2r = 2 \times 4^2 \times 1 = 32$N

23. Centripetal acceleration,

$a_c = \frac{V^2}{r} = \frac{30^2}{500} = 1.8\ \text{m/s}^2$

Tangential acceleration $a_t = 2\ \text{m/s}^2$

$a = \sqrt{a_c^2 + a_t^2} = 2.7\ \text{m/s}^2$

25. Max^m time period = $2\pi\sqrt{\frac{r}{g}} = 2\pi \times \sqrt{\frac{4}{10}}$ = 4sec

26. $\tan\theta = \frac{V^2}{rg} = \frac{2\pi c}{t^2 g}$ $\left[\because V = \frac{c}{t} = \frac{2\pi r}{t}\right]$

$\tan\theta = \left(\frac{2 \times \frac{22}{7} \times 34.3}{22 \times 9.8}\right) = 1$

$\therefore \theta = 45°$

27. $\frac{h}{L} = \frac{V^2}{rg}$ where, h = height raised

L = distance between rails

$h = \frac{\left(48 \times \frac{5}{18}\right)^2}{400 \times 10} \times 1$

$h = 4.5$cm

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